

ECM402 Lab 2: Signal Classification (Track A: ECG Arrhythmia) & (Track B: Spoken Digit Classification)

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1 Mathematical Formulas Methodology

1.1 Data Preprocessing

Feature Standardization: To prevent scale-dependent models (like SVMs) from being dominated by large-magnitude features, the training data was scaled to have a mean of 0 and a standard deviation of 1[cite: 1, 2].

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \quad (1)$$

1.2 Logistic Regression

Hypothesis (Sigmoid Function): The probability of a sample belonging to class 1 is calculated using the sigmoid activation function[cite: 1, 2]:

$$\hat{y}_i = \sigma(w^T x_i + b) = \frac{1}{1 + e^{-(w^T x_i + b)}} \quad (2)$$

Objective Function (Regularized Log-Loss): Minimizing Binary Cross-Entropy with l_2 regularization[cite: 1, 2]:

$$L(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{\alpha}{2n} \|w\|_2^2 \quad (3)$$

Batch Gradient Descent Updates:

$$\nabla_w L = \frac{1}{n} X^T (\hat{y} - y) + \frac{\alpha}{n} w \quad (4)$$

$$\nabla_b L = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad (5)$$

1.3 Support Vector Machines (SVM)

Linear SVM Margins: The geometric margins defining the support vectors are[cite: 1]:

$$w^T x + b = \pm 1 \quad (6)$$

Radial Basis Function (RBF) Kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (7)$$

Non-Linear Decision Function:

$$f(x) = \sum_i \alpha_i y_i K(x_i, x) + b = 0 \quad (8)$$

1.4 Random Forest & Ensembles

Random Forest Theory: Random Forests are an ensemble learning method that operates by constructing a multitude of decision trees. It combines "bagging" (bootstrap aggregating) with random feature selection at each node split. By training many deep trees on random subsets of the data and features, the ensemble decorrelates the individual trees, which significantly reduces the overall model variance and prevents overfitting[cite: 1].

Gini Impurity (Node Splitting & Feature Importance): The decision trees split nodes by minimizing the Gini Impurity, defined as[cite: 1]:

$$I_G(p) = 1 - \sum_{k=1}^K p_k^2 \quad (9)$$

where K is the total number of classes, and p_k is the fraction of items belonging to class k at a given node. This is also used to calculate the Mean Decrease in Impurity (MDI) for feature importance.

Majority Voting (Classification): For a given input feature vector x , let $\hat{C}_b(x)$ be the class predicted by the b -th tree out of B total trees. The final Random Forest prediction (and the Hard-Voting Ensemble prediction) is the mode (majority vote)[cite: 1, 2]:

$$\hat{y}_{RF} = \text{mode}\{\hat{C}_1(x), \hat{C}_2(x), \dots, \hat{C}_B(x)\} \quad (10)$$

1.5 Evaluation Metrics

Macro-F1 Score: Evaluates performance accurately under potential class imbalances by calculating the unweighted mean of the F1 scores across all classes[cite: 1].

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

2 Dataset Class Distribution & Analysis

Figure 1 illustrates the class frequency distribution across the AAMI superclasses (N, V, S, F, Q) for Track A (ECG Arrhythmia Database). The dataset exhibits a notable class imbalance, with normal beats (N) dominating the total count.

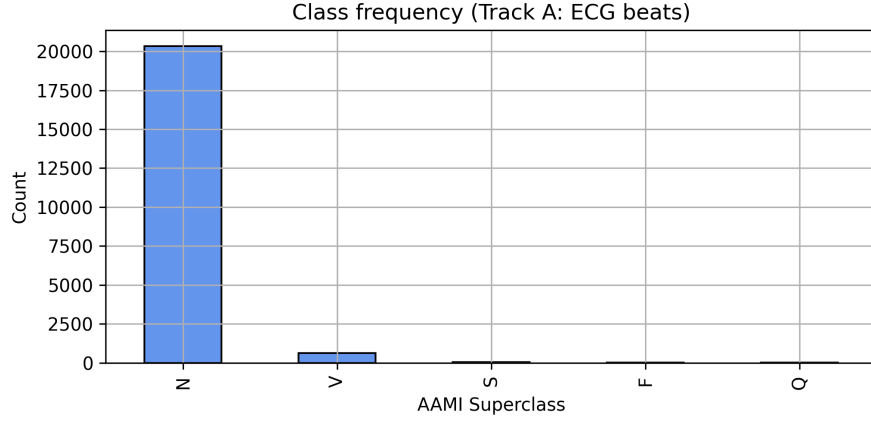


Figure 1: Class frequency distribution for Track A (ECG beats).

3 Model Training & Hyperparameter Tuning

3.1 Multiclass Logistic Regression Confusion Matrix

Figure 2 shows the training confusion matrix for the multiclass logistic regression model across the AAMI superclasses.

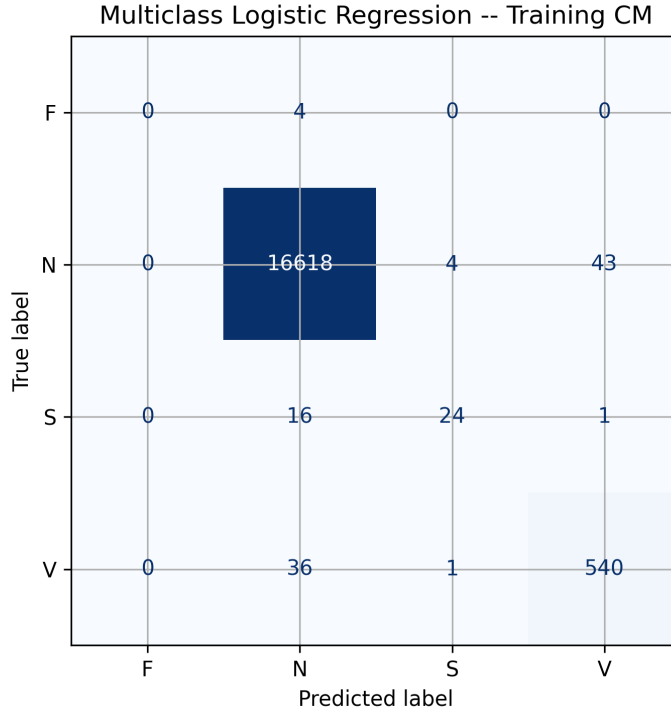


Figure 2: Multiclass Logistic Regression training confusion matrix.

3.2 RBF-SVM Hyperparameter Grid Search

Figure 3 displays the 5-fold grouped cross-validation macro-F1 heatmap over the (C, γ) parameter grid for the RBF-kernel Support Vector Machine.

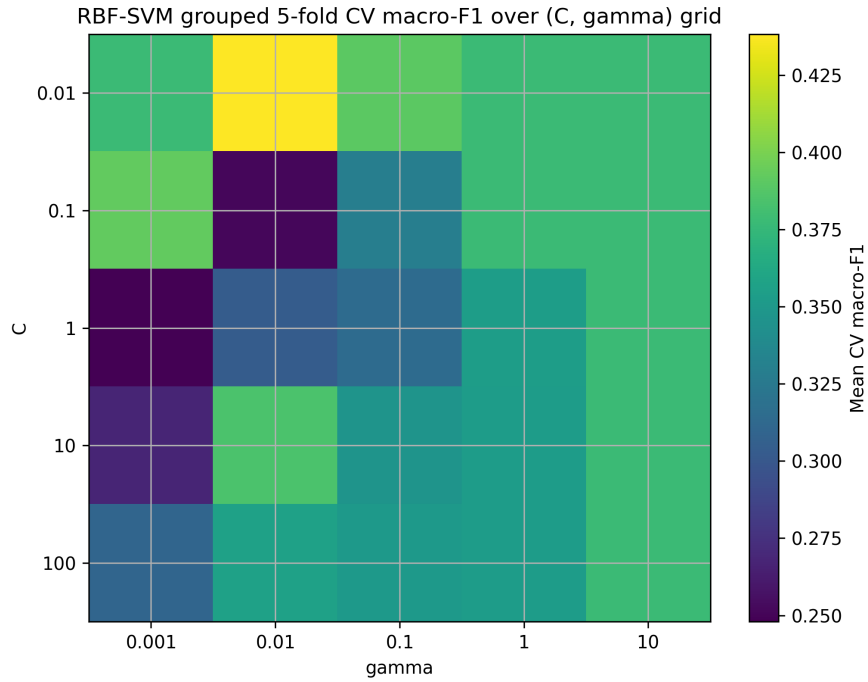


Figure 3: RBF-SVM grouped 5-fold CV macro-F1 heatmap over (C, γ) .

4 Error Analysis: Misclassified ECG Beats

During model evaluation, specific beats were misclassified across similar morphological classes. Figures 4 to 6 display the raw waveform segments of individual misclassified heartbeats.

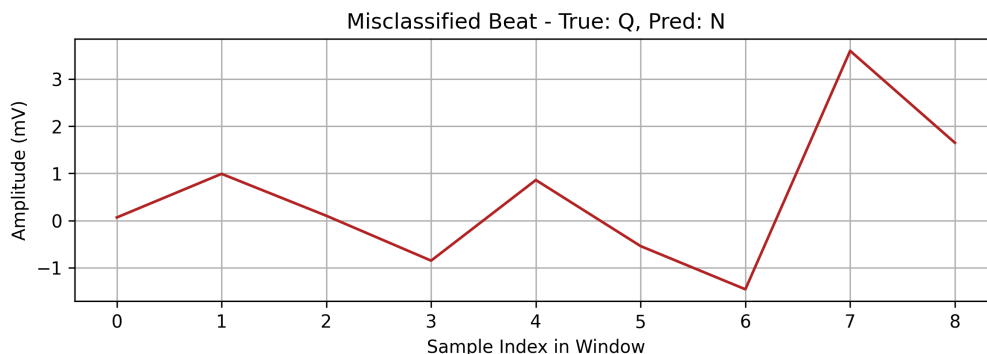


Figure 4: Misclassified Beat 1: True Label Q, Predicted Label N.

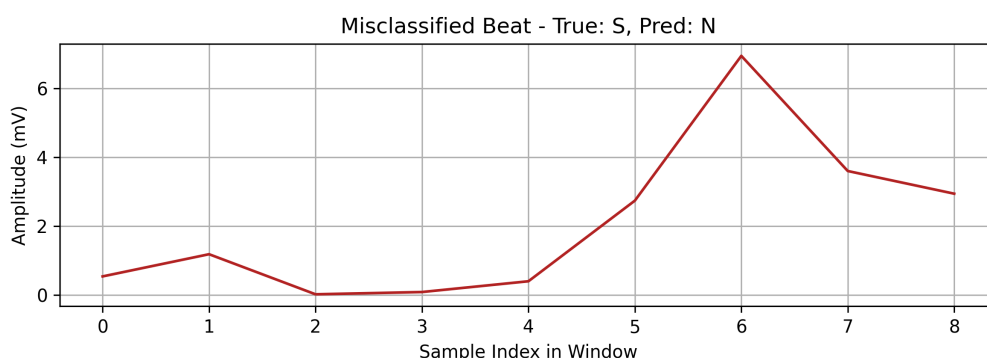


Figure 5: Misclassified Beat 2: True Label S, Predicted Label N.

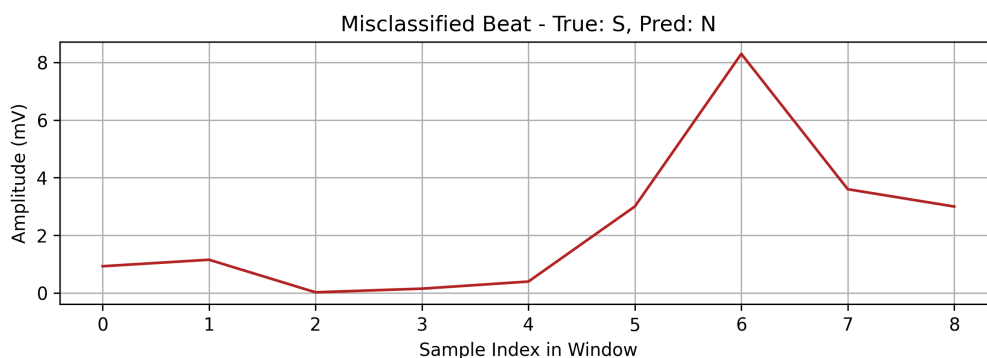


Figure 6: Misclassified Beat 3: True Label S, Predicted Label N.

4.1 Data Preprocessing

Feature Standardization: To prevent scale-dependent models (like SVMs) from being dominated by large-magnitude features, the training data was scaled to have a mean of

0 and a standard deviation of 1.

$$x' = \frac{x - \mu_{train}}{\sigma_{train}} \quad (12)$$

4.2 Logistic Regression

Hypothesis (Sigmoid Function): The probability of a sample belonging to class 1 is calculated using the sigmoid activation function:

$$\hat{y}_i = \sigma(w^T x_i + b) = \frac{1}{1 + e^{-(w^T x_i + b)}} \quad (13)$$

Objective Function (Regularized Log-Loss): Minimizing Binary Cross-Entropy with l_2 regularization:

$$L(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{\alpha}{2n} \|w\|_2^2 \quad (14)$$

Batch Gradient Descent Updates:

$$\nabla_w L = \frac{1}{n} X^T (\hat{y} - y) + \frac{\alpha}{n} w \quad (15)$$

$$\nabla_b L = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad (16)$$

4.3 Support Vector Machines (SVM)

Linear SVM Margins: The geometric margins defining the support vectors are:

$$w^T x + b = \pm 1 \quad (17)$$

Radial Basis Function (RBF) Kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (18)$$

Non-Linear Decision Function:

$$f(x) = \sum_i \alpha_i y_i K(x_i, x) + b = 0 \quad (19)$$

4.4 Random Forest & Ensembles

Random Forest Theory: Random Forests are an ensemble learning method that operates by constructing a multitude of decision trees. It combines "bagging" (bootstrap aggregating) with random feature selection at each node split. By training many deep trees on random subsets of the data and features, the ensemble decorrelates the individual trees, which significantly reduces the overall model variance and prevents overfitting.

Gini Impurity (Node Splitting & Feature Importance): The decision trees split nodes by minimizing the Gini Impurity, defined as:

$$I_G(p) = 1 - \sum_{k=1}^K p_k^2 \quad (20)$$

where K is the total number of classes, and p_k is the fraction of items belonging to class k at a given node. This is also used to calculate the Mean Decrease in Impurity (MDI) for feature importance.

Majority Voting (Classification): For a given input feature vector x , let $\hat{C}_b(x)$ be the class predicted by the b -th tree out of B total trees. The final Random Forest prediction (and the Hard-Voting Ensemble prediction) is the mode (majority vote):

$$\hat{y}_{RF} = \text{mode}\{\hat{C}_1(x), \hat{C}_2(x), \dots, \hat{C}_B(x)\} \quad (21)$$

4.5 Evaluation Metrics

Macro-F1 Score: Evaluates performance accurately under potential class imbalances by calculating the unweighted mean of the F1 scores across all classes.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

5 Model Comparison: RBF SVM vs. Random Forest

An exploratory paired t-test was conducted on the 5-fold grouped cross-validation scores to compare the RBF SVM and Random Forest models.

Metric	Value
Per-fold differences	[0.0649, −0.0897, 0.0911, −0.0811, −0.1405]
Mean difference	−0.0311
t-statistic	−0.6776
p-value	0.5352

Table 1: Paired t-test results (Not statistically significant at $p < 0.05$)

6 Final Test Set Evaluation

The final models were evaluated on the entirely untouched held-out test set (Speaker: `yweweler`).

Model	Accuracy	Macro-F1
RBF SVM	0.5960	0.5824
Voting Ensemble (Hard)	0.6560	0.6509

Table 2: Final test set metrics

7 Confusion Matrix

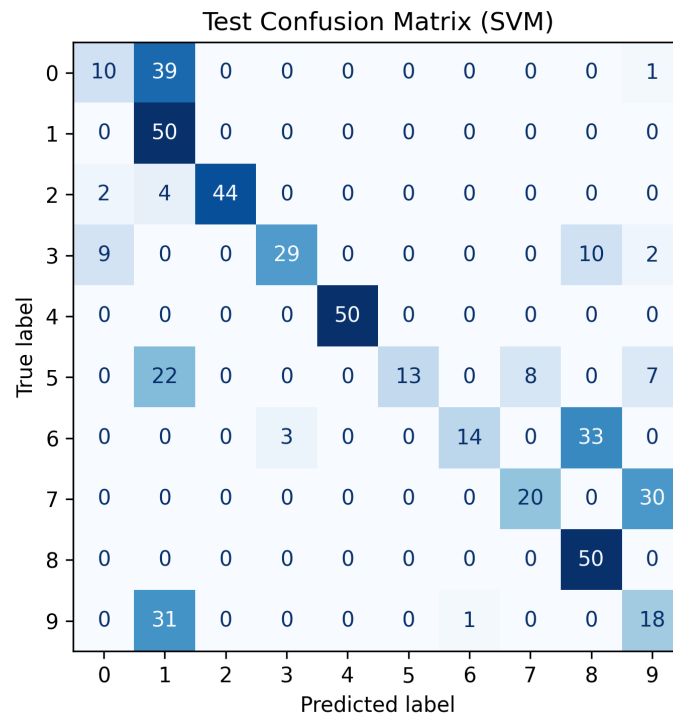


Figure 7: Test Confusion Matrix for the RBF SVM model.

8 Error Analysis

During the final test set evaluation, the RBF SVM model misclassified a total of **202 samples**. The following figures display the waveform and spectrogram for 5 misclassified test samples where the true spoken digit was **0** but the model incorrectly predicted **1**.

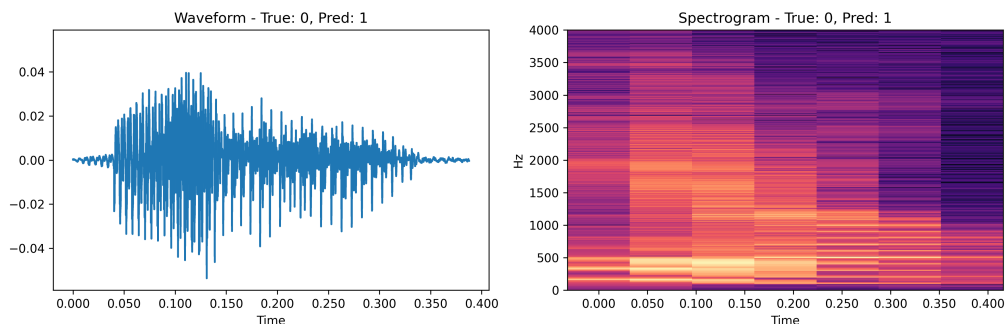


Figure 8: Error Analysis Example 1: True Label 0, Predicted Label 1

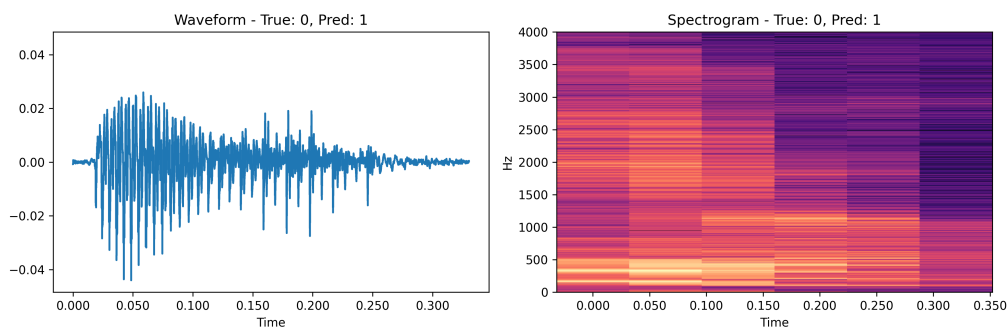


Figure 9: Error Analysis Example 2: True Label 0, Predicted Label 1

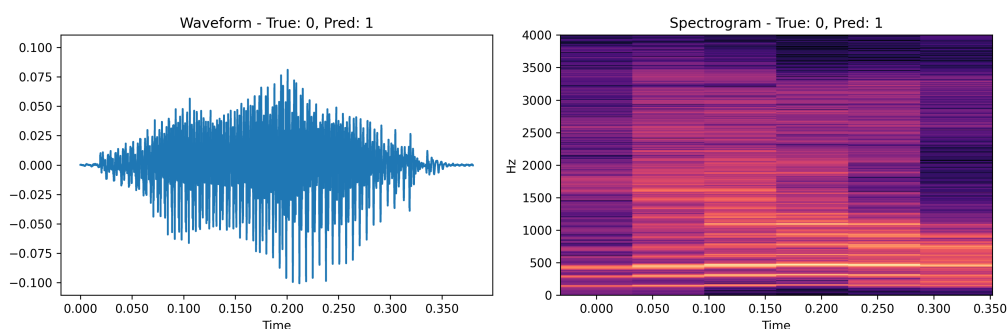


Figure 10: Error Analysis Example 3: True Label 0, Predicted Label 1

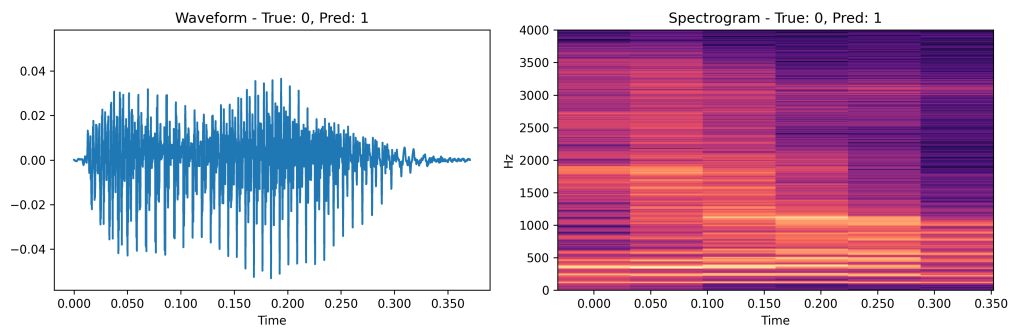


Figure 11: Error Analysis Example 4: True Label 0, Predicted Label 1

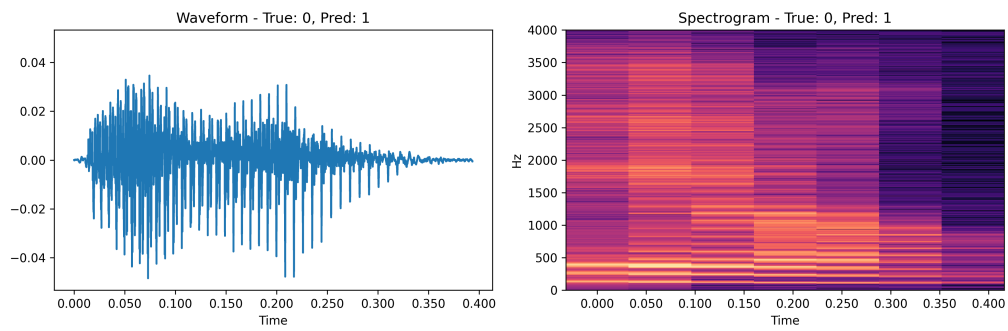


Figure 12: Error Analysis Example 5: True Label 0, Predicted Label 1